[osunlp/MagicBrush · Datasets at Hugging Face](#)

Magicbrush: A manually annotated dataset for instruction-guided image editing [PDF] neurips.cc
K Zhang, L Mo, W Chen, H Sun… - Advances in Neural …, 2023 - proceedings.neurips.cc
… To address this issue, we introduce **MagicBrush**, the first large-… **MagicBrush** comprises over 10K manually annotated triplets (… We fine-tune InstructPix2Pix on **MagicBrush** and show that …
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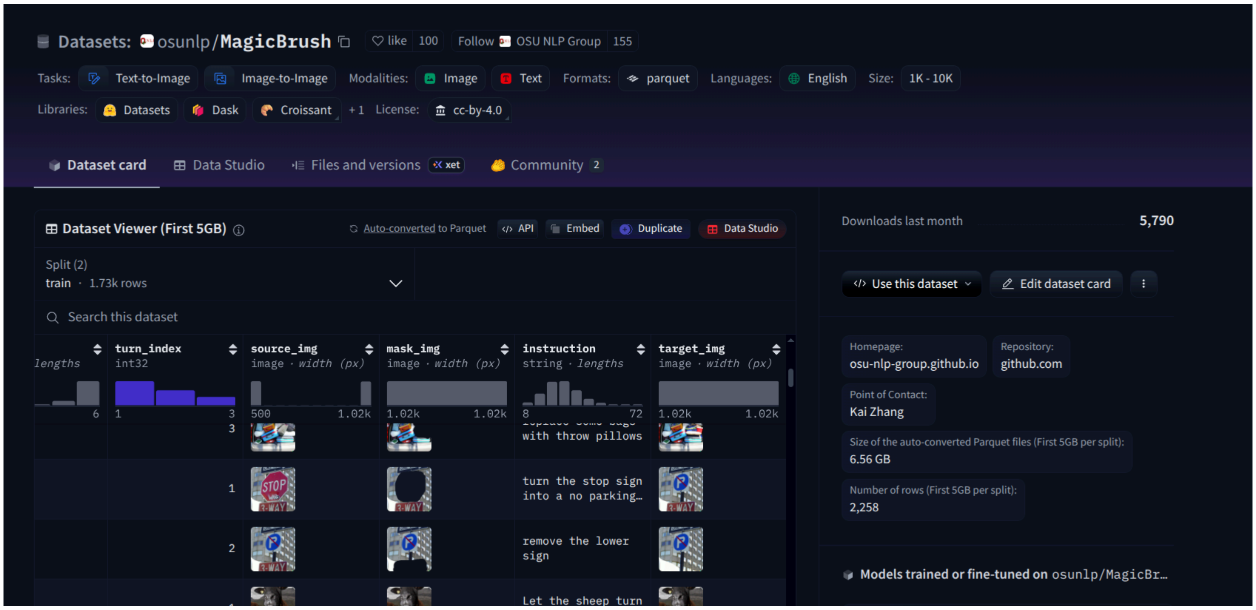

Project Overview

We study **instruction-following failures in text-guided image editing** by introducing human-annotated **instruction adherence labels** and a structured **error taxonomy** over image-editing triplets:

(original image, edit prompt, edited image)

For each triplet, we collect explicit human judgments of whether the edit prompt is correctly reflected in the edited image, then analyze how **prompt properties** and **image characteristics** correlate with success and failure.

Data Format

Each sample is an **input triplet**:

- 🖼️ **Original Image**

- 📝 **Edit Prompt (Instruction)**
- 🖼️ **Edited Image (Model Output)**

Source dataset: MagicBrush (as the base triplet dataset—osunlp/MagicBrush · Datasets at Hugging Face).
We extend it by adding human annotations for instruction adherence and error types.

Human Annotation Protocol

Step 1: Instruction Adherence (Ordinal Label)

Annotators answer:

- **Yes (Success):** The prompt is fully and correctly reflected in the edited image.
- **Partial:** The edit reflects some intent, but important parts are missing or incorrect.
- **No (Failure):** The prompt is not followed (or the result contradicts the prompt).

This gives an explicit, human-grounded measure of **instruction fulfillment**.

Step 2: Error Type Classification (Multi-Label)

If the adherence label is **Partial** or **No**, annotators select one or more **error categories** that best explain what went wrong.

Annotators may select **multiple** categories when necessary.

Category	Meaning
❌ **Wrong Object**	Edited the wrong object
❌ **Missing Object**	Object should be added but isn't
❌ **Extra Object**	Added something not requested
❌ **Wrong Attribute**	Color / size / texture incorrect

❌ **Spatial Error**	Wrong location / relation
❌ **Style Mismatch**	Style not as requested
❌ **Over-editing**	Changed things that shouldn't change
❌ **Under-editing**	Edit is too weak / barely visible
❌ **Artifact / Quality Issue**	Visual artifacts
❌ **Ambiguous Prompt**	Prompt itself unclear

End-to-End Workflow

Pipeline:

Original Image
→ Edit Prompt
→ Edited Image
→ Human Annotation (Success / Partial / Failure)
→ Error Type Classification (for Partial/Failure)
→ Error Taxonomy + Statistical Analysis

Analysis: Correlating Prompt Properties with Failure Modes

We analyze **why edits fail** and how failure relates to:

- **Prompt properties:** ambiguity, number of requested changes, number of objects, attribute complexity (color/texture/style), spatial relations, specificity of references ("the man on the left"), etc.

- **Image properties:** whether the referenced object exists, clutter, multiple similar objects, small objects, occlusion, etc.

This yields systematic insights such as which prompt patterns most strongly predict:

- over-editing (collateral changes),
- wrong-object edits,
- under-edits,
- style mismatches, and more.

Optional Deep Learning Component: Automatic Error Prediction / Scoring

We also explore training an automatic predictor using the annotated data:

A) Output Evaluation Model

Train a classifier/scorer that predicts adherence and error types from the full triplet:

(original image + edit prompt + edited image) → predicted success/partial/failure and error category(ies)

Use cases:

- automatic quality estimation,
- selecting the best output among multiple candidates,
- producing model-specific error statistics at scale.

B) Difficulty Prediction

Predict whether a prompt is likely to fail **before** editing:

(original image + edit prompt) → predicted failure risk / expected error type

Use cases:

- prompt warnings (“high risk of over-editing”),
- targeted prompt rewrite suggestions.

(We are planning to add this but not sure whether it will be feasible or not)